

Computer Networks

Lab Manual

Experiment -1

Aim : To manage and troubleshoot communication between computers in a network.

Commands and its functions:

- i) ipconfig – It displays the IP configuration of a computer.
- Displays the **IP address** of the device
 - Shows the **Subnet Mask**
 - Shows the **Default Gateway**
 - Displays **DNS server information**
 - Helps **troubleshoot network connectivity issues**
 - Can **release and renew IP addresses** assigned by a DHCP server

```
Microsoft Windows [Version 10.0.14393]
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C:\Users\vignan>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a014:218a:3de:8bc7%16
    IPv4 Address. . . . . : 172.21.33.75
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.21.33.1

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3488:abf2:2b1e:1002%8
    IPv4 Address. . . . . : 192.168.174.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::f8fc:a1f:5c11:12f2%12
    IPv4 Address. . . . . : 192.168.232.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Tunnel adapter Local Area Connection* 10:

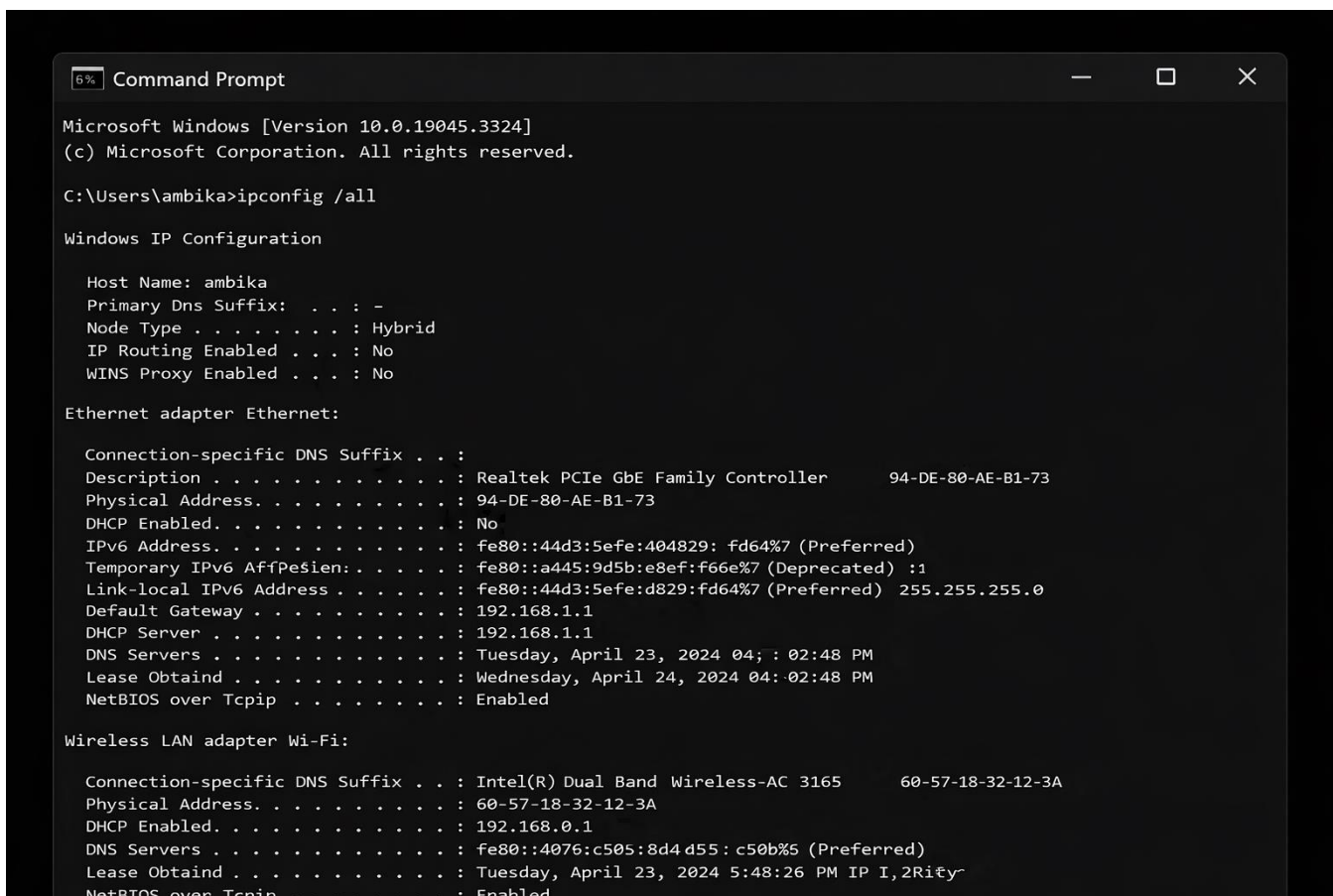
    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:0:14c9:d206:431:9a1:5f44:56f9
    Link-local IPv6 Address . . . . . : fe80::431:9a1:5f44:56f9%9
    Default Gateway . . . . . : ::

Tunnel adapter isatap.{2DEA074A-285C-4261-838A-979F53641C24}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

ii) `ipconfig /all` – Shows all the connections connected to the device.

- Displays the **host name** of the computer
- Shows the **MAC (Physical) address** of each network adapter
- Displays the **IPv4 and IPv6 addresses**
- Shows the **Subnet Mask**
- Displays the **Default Gateway**
- Shows **DNS server addresses**
- Displays **DHCP status** (enabled/disabled)
- Shows **DHCP server address**
- Displays **lease obtained and lease expiry time**
- Shows **NetBIOS over TCP/IP status**



```
Microsoft Windows [Version 10.0.19045.3324]
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C:\Users\ambika>ipconfig /all

Windows IP Configuration

Host Name: ambika
Primary Dns Suffix: . . . : -
Node Type . . . . . : Hybrid
IP Routing Enabled . . . : No
WINS Proxy Enabled . . . : No

Ethernet adapter Ethernet:

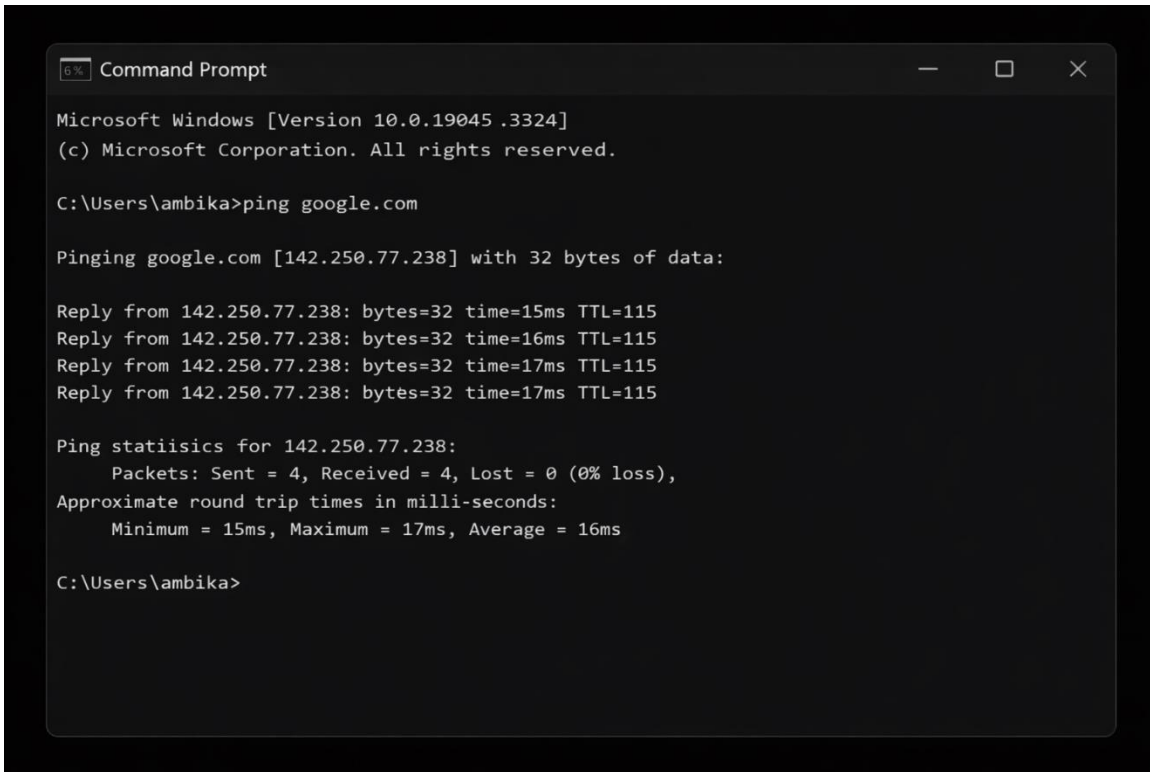
Connection-specific DNS Suffix . . . :
Description . . . . . : Realtek PCIe GbE Family Controller 94-DE-80-AE-B1-73
Physical Address. . . . . : 94-DE-80-AE-B1-73
DHCP Enabled. . . . . : No
IPv6 Address. . . . . : fe80::44d3:5efe:404829:fd64%7 (Preferred)
Temporary IPv6 Address. . . . . : fe80::a445:9d5b:e8ef:f66e%7 (Deprecated) :1
Link-local IPv6 Address . . . . . : fe80::44d3:5efe:d829:fd64%7 (Preferred) 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DNS Servers . . . . . : Tuesday, April 23, 2024 04:02:48 PM
Lease Obtained . . . . . : Wednesday, April 24, 2024 04:02:48 PM
NetBIOS over Tcpip . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . . : Intel(R) Dual Band Wireless-AC 3165 60-57-18-32-12-3A
Physical Address. . . . . : 60-57-18-32-12-3A
DHCP Enabled. . . . . : 192.168.0.1
DNS Servers . . . . . : fe80::4076:c505:8d4d55:c50b%5 (Preferred)
Lease Obtained . . . . . : Tuesday, April 23, 2024 5:48:26 PM IP I,2Rityr
NetBIOS over Tcpip . . . . . : Enabled
```

iii) Ping – Checking the availability of IP address in the network.

- Checks whether a **remote host is reachable**
- Tests the **network connection** between devices
- Measures **response time (latency)**
- Detects **packet loss**
- Helps **troubleshoot network problems**

A screenshot of a Windows Command Prompt window. The title bar says "Command Prompt" with a 5% zoom indicator. The window content shows the following text:

```
Microsoft Windows [Version 10.0.19045.3324]
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C:\Users\ambika>ping google.com

Pinging google.com [142.250.77.238] with 32 bytes of data:

Reply from 142.250.77.238: bytes=32 time=15ms TTL=115
Reply from 142.250.77.238: bytes=32 time=16ms TTL=115
Reply from 142.250.77.238: bytes=32 time=17ms TTL=115
Reply from 142.250.77.238: bytes=32 time=17ms TTL=115

Ping statistics for 142.250.77.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 17ms, Average = 16ms

C:\Users\ambika>
```

iv) traceroute – Finds the route by the data packets from one computer to another.

- Shows the **route (path)** packets take to reach a destination
- Displays each **hop** (router) along the path
- Measures **time taken** to reach each hop
- Helps **identify where network delays or failures occur**
- Assists in **troubleshooting network problems**

v) nslookup – Gives the DNS Information

- Finds the **IP address of a domain name**
- Finds the **domain name from an IP address**
- Checks whether a **DNS server is working properly**
- Displays **DNS records** (such as A, MX, CNAME, etc.)
- Helps **troubleshoot DNS-related problems**

vi) netstat – a – Shows all the connections available in the network.

netstat –n – Shows only TCP connections

netstat –r – Routing table information

- Shows **active network connections**
- Displays **listening ports**
- Shows **IP addresses and port numbers**
- Displays **protocols** in use (TCP, UDP)
- Shows **network interface statistics**
- Helps **detect suspicious or unwanted connections**

- Assists in **network troubleshooting**

```
Command Prompt
Microsoft Windows [Version 10.0.14393]
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C:\Users\ambika>tracert google.com

Tracing route to google.com [142.250.192.206] over a maximum of 30 hops:

  0 ms  1 ms  2 ms  14.98.182.1
  1 ms  1 ms  1 ms  115.110.208.129.static-noida.vsnl.net.in [115.110.208.129]
  3 ms  1 ms  1 ms  59.97.144.33.static-mumbai.vsnl.net.in [59.97.144.33]
  1 ms  2 ms  1 ms  59.163.121.138.static-chennai.vsnl.net.in [59.163.121.138]
  5 ms  2 ms  2 ms  59.163.109.214.static-chennai.jetnl.vsnl.net.in [59.163.109.214]
  6 ms  *    *    *
  7 ms  *    *    * Request timed out.
  7 ms  261 ms  61 ms  222.165.241.182 / 72.143.220.110
  8 ms  261 ms  39 ms  209.85.245.141
 10 ms  31 ms  39 ms  142.251.56.55
 11 ms  260 ms  261 ms  maa03s24-in-f1f.1f05.net [142.250.56.55]
 11 ms  260 ms  261 ms  maa03s24-in-f14.1e100.net [142.250.192.206]

Trace complete.

C:\Users\ambika>
```

```
Command Prompt
Microsoft Windows [Version 10.0.14393]
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C:\Users\ambika>nslookup google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4007:836::200e
          142.250.77.142

C:\Users\ambika>
```

```
Command Prompt
Microsoft Windows [Version 10.0.14393]
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C:\Users\ambika>netstat -a

Active Connections
Proto Local Address           Foreign Address         State
TCP    0.0.0.0:135              0.0.0.0:0               LISTENING
TCP    0.0.0.0:445              0.0.0.0:0               LISTENING
TCP    0.0.0.0:5040             0.0.0.0:0               LISTENING
TCP    127.0.0.1:5354           0.0.0.0:0               LISTENING
TCP    192.168.1.72:139        0.0.0.0:0               LISTENING
TCP    192.168.1.72:49215      192.168.1.1:53          ESTABLISHED
TCP    192.168.1.72:49216      142.250.77.142:443      ESTABLISHED
TCP    192.168.1.72:49235      13.107.42.16:443        ESTABLISHED
UDP    0.0.0.0:123             *:*                     LISTENING
UDP    0.0.0.0:5353            *:*                     LISTENING
UDP    192.168.1.72:137       *:*                     LISTENING

C:\Users\ambika>netstat -n

Active TCP Connections (Numeric Only)
Proto Local Address           Foreign Address         State
TCP    192.168.1.72:49215      192.168.1.1:53          ESTABLISHED
TCP    192.168.1.72:49216      142.250.77.142:443      ESTABLISHED
TCP    192.168.1.72:49335      13.107.42.16:443        ESTABLISHED
TCP    127.0.0.1:5354          0.0.0.0:0               LISTENING
TCP    0.0.0.0:123             0.0.0.0:0               LISTENING

C:\Users\ambika>netstat -r

IPv4 Route Table
=====
Network Destination     Netmask          Gateway          Interface        Metric
0.0.0.0                  0.0.0.0          192.168.1.1      192.168.1.23     25
42..0.0.0                255.0.0.0        On-link          127.0.0.1         331
222.168.1.0              253.250.473.0    On-link          122.168.1.72     201
324.0.0.0                310.0.0.0        On-link          222.168.1.23     281
222.256.475.255          255.282.255.255  On-link          192.168.1.22     281
```

Result:

The network diagnostic commands **ipconfig**, **ipconfig /all**, **ping**, **tracert**, **nslookup**, and **netstat** were successfully executed using the Windows Command Prompt. The **ipconfig** command displayed the system's IP configuration including IPv4 address, subnet mask, and default gateway. The **ipconfig /all** command showed detailed network information such as host name, MAC address, DNS server, and DHCP details. The **ping** command verified network connectivity and measured response time between the local computer and a remote host. The **tracert** command identified the path taken by data packets through multiple routers to reach the destination. The **nslookup** command provided DNS information by resolving the domain name into its corresponding IP address. The **netstat** command displayed active network connections, listening ports, and routing table information. Thus, these commands helped in analyzing network configuration, connectivity, and troubleshooting network issues.

Experiment -2

Aim : To configure a computer system as a **host** and allow **other systems (clients)** on the same network to access it.

Procedure :

- Connect the **host system** and **client systems** to the **same network** (LAN or Wi-Fi).
- On the host system, open Command Prompt and note the **IP address** using the ipconfig command.
- Enable the required **network service** on the host system (such as file sharing or a basic service for access).
- Ensure the host system allows network access (network discovery enabled).
- From the client system, verify connectivity to the host using the ping command with the host IP address.
- Access the host system from the client using the **host IP address** through the appropriate method (network location or browser, as applicable).
- Observe whether the client system can successfully communicate with and access the host.

Output :

```
Command Prompt - python server.py
Microsoft Windows [Version 10.0.14393]
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C:\Users\vignan>cd desktop
C:\Users\vignan\Desktop>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::193a:736f:316c:6b95%5
    IPv4 Address. . . . . : 172.22.36.21
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.22.36.1

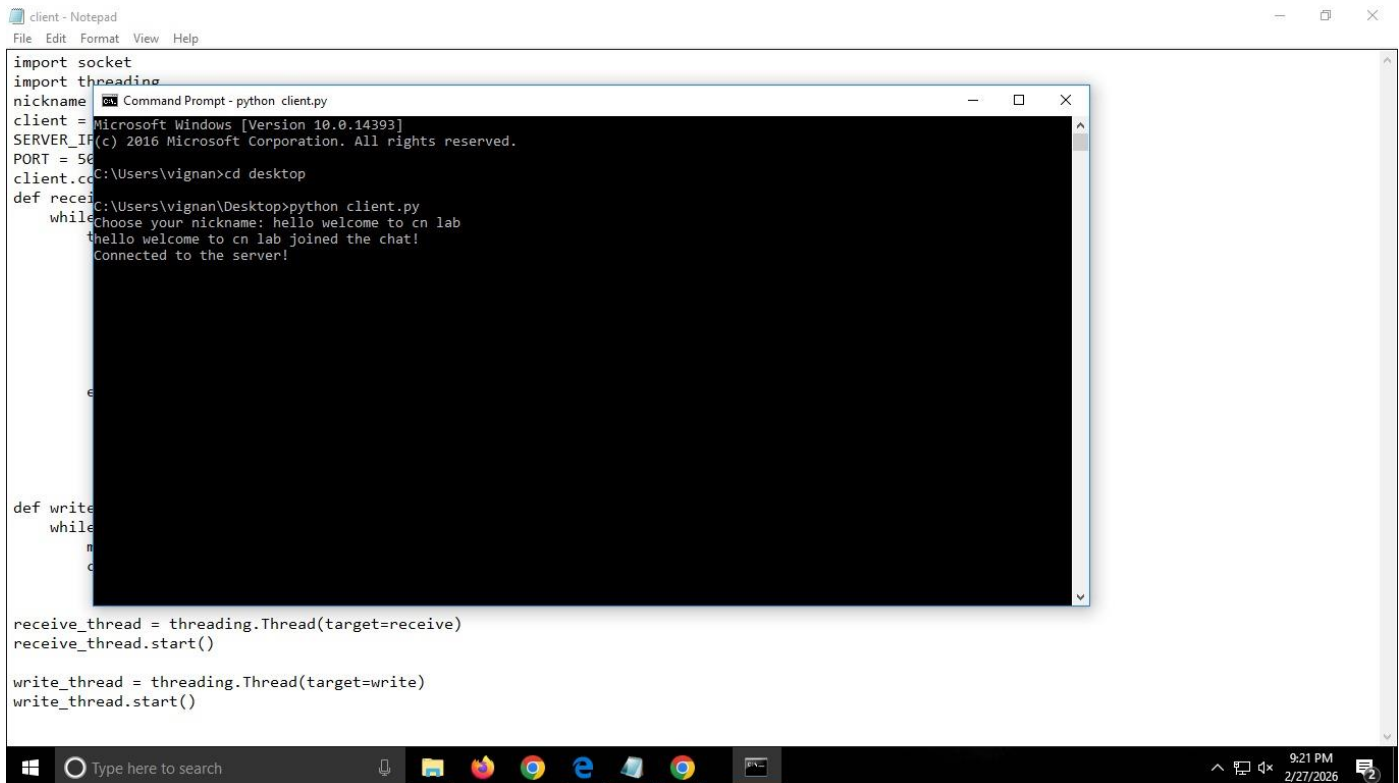
Tunnel adapter isatap.{3FF45C13-F6A5-45BF-A1CC-6C42EF33E768}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter Teredo Tunneling Pseudo-Interface:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:0:14c9:d206:1c66:3162:8a3f:f60a
    Link-local IPv6 Address . . . . . : fe80::1c66:3162:8a3f:f60a%9
    Default Gateway . . . . . : ::

C:\Users\vignan\Desktop>python server.py
Server started. Waiting for connections...
Connected with ('172.22.36.52', 6754)
Nickname is hello welcome to cn lab
```

The screenshot shows a Windows 10 desktop. A Notepad window titled 'client - Notepad' contains the following Python code:

```
import socket
import threading
nickname = ''
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
SERVER_IP = '192.168.1.100'
PORT = 5000
client.connect((SERVER_IP, PORT))
def receive():
    while True:
        message = client.recv(1024)
        print(message)
def write():
    while True:
        message = input(' ')
        client.send(message.encode())
receive_thread = threading.Thread(target=receive)
receive_thread.start()
write_thread = threading.Thread(target=write)
write_thread.start()
```

Overlaid on the Notepad window is a Command Prompt window titled 'Command Prompt - python client.py'. It shows the execution of the script:

```
C:\Users\vignan>cd desktop
C:\Users\vignan\Desktop>python client.py
Choose your nickname: hello welcome to cn lab
hello welcome to cn lab joined the chat!
Connected to the server!
```

The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system tray on the right indicates the time is 9:21 PM on 2/27/2026.

Result -

The computer system was successfully configured as a **host**, and other systems on the same network were able to access it. The host IP address was identified using the **ipconfig** command, and network services such as **network discovery** and **file sharing** were enabled. The client system verified connectivity to the host by using the **ping** command with the host IP address, which confirmed successful communication between the systems. The client was then able to access the host system through the network using the host IP address. This experiment demonstrated successful **host configuration and communication between host and client systems within the same network**

Experiment – 3

Aim : Establishing the LAN connection between the systems.

Procedure :

Step 1: Place Devices & Connect

1. Add Devices: Drag PCs (End Devices) and a Switch (or Hub) onto the workspace.
2. Connect: Use the Copper Straight-Through cable (lightning bolt icon).
3. Click the cable, select a PC's FastEthernet0 port, then click the switch and choose an available port. Repeat for all PCs.

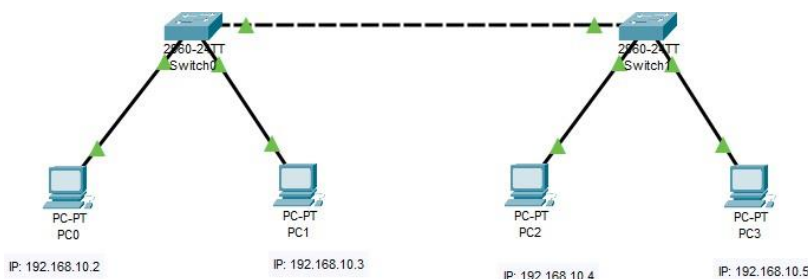
Step 2: Configure IP Addresses

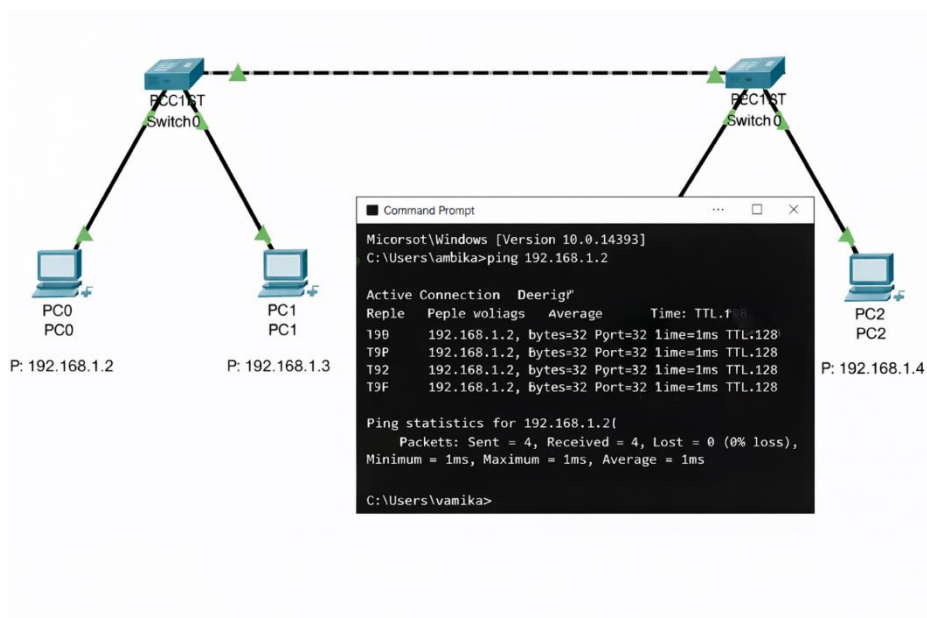
1. Select PC: Click a PC, go to the Desktop tab, then IP Configuration.
2. Set Static IP: Choose Static, enter an IP address (e.g., 192.168.1.1), Subnet Mask (e.g., 255.255.255.0), and leave Default Gateway blank for a simple LAN.
3. Repeat: Assign unique IPs to other PCs in the *same network range* (e.g., 192.168.1.2 , 192.168.1.3).

Step 3: Test Connectivity

1. Open Command Prompt: Click a PC, go to Desktop, then Command Prompt.
2. Ping: Type `ping <IP_address_of_another_PC>` (e.g., `ping 192.168.1.2`) and press Enter.
3. Verify: Successful replies (0% packet loss) confirm the LAN connection is working.

Output:





Result :

The LAN connection between the systems was successfully established using a switch and copper straight-through cables. Each PC was assigned a unique IP address within the same network range, and connectivity between the systems was tested using the **ping** command in the Command Prompt. The successful reply messages with **0% packet loss** confirmed that the PCs were able to communicate with each other, indicating that the LAN network was configured and functioning correctly.

Experiment – 4

Aim : Configuration of the Router.

Procedure :

Step 1: Place Devices & Connect

1. Add Devices:
Drag PCs (End Devices), two Switches, and one Router onto the workspace.
2. Connect PCs to Switches:
 - Use the Copper Straight-Through cable (lightning bolt icon).
 - Click the cable → select PC FastEthernet0 → click the Switch → choose an available FastEthernet port.
 - Repeat for all PCs (PCs can be split between the two switches).
3. Connect Switches to Router:
 - Use Copper Straight-Through cables.
 - Connect Switch 1 to Router FastEthernet0/0.
 - Connect Switch 2 to Router FastEthernet0/1.

Step 2: Configure IP Addresses on Router

1. Select Router: Click the router and go to the Config tab.
2. Configure Interfaces:
 - FastEthernet0/0
 - IP Address: 192.168.1.1
 - Subnet Mask: 255.255.255.0
 - FastEthernet0/1
 - IP Address: 192.168.2.1
 - Subnet Mask: 255.255.255.0
3. Make sure both interfaces are ON.

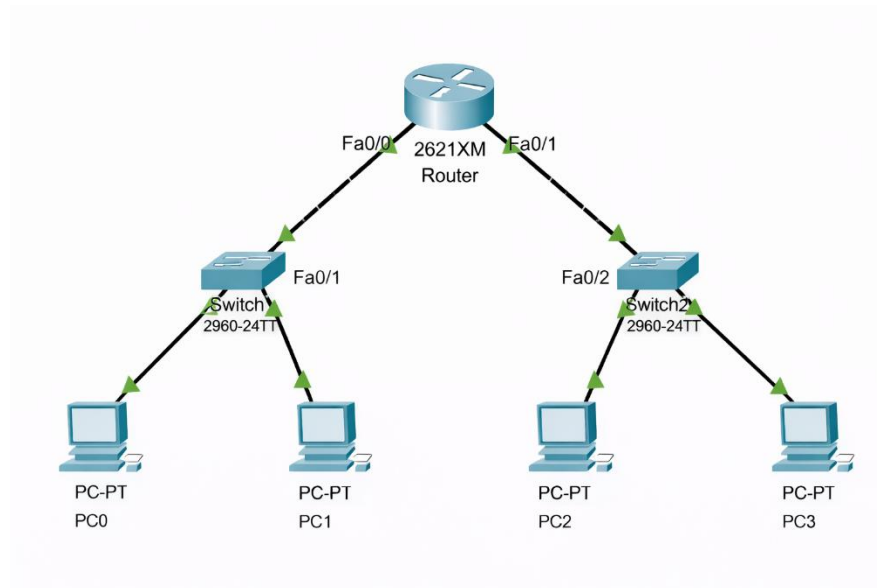
Step 3: Configure IP Addresses on PCs

1. Select a PC: Click a PC → Desktop tab → IP Configuration.
2. Set Static IP:
 - For PCs on Switch 1:
 - IP Address: 192.168.3.2, 192.168.3.3, etc.
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.3.1
 - For PCs on Switch 2:
 - IP Address: 192.168.4.2, 192.168.4.3, etc.
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.4.1
3. Repeat for all PCs, ensuring each has a unique IP.

Step 4: Test Connectivity

1. Open Command Prompt:
Click a PC → Desktop → Command Prompt.
2. Ping Same Network:
3. ping <IP_of_PC_on_same_switch>
4. Ping Across Networks:
5. ping <IP_of_PC_on_other_switch>
6. Verify:
Successful replies (0% packet loss) confirm that the router is correctly connecting both switches.

Output :



```
Physical Config Desktop Programming Attributes
Command Prompt

C:\>ping 192.168.16.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time<1ms TTL=126
Reply from 192.168.3.3: bytes=32 time<1ms TTL=126
Reply from 192.168.3.3: bytes=32 time<1ms TTL=126
Reply from 192.168.3.3: bytes=32 time<1ms TTL=126

C:\>stats statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli=seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.4.2

Pinging 192.168.4.3 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time<1ms TTL=126
Reply from 192.168.4.2: bytes=32 time<1ms TTL=126
Reply from 192.168.4.2: bytes=32 time<1ms TTL=126
C >|
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
✓	Successful	PC0	PC1	ICMP	Light Blue	16 (ecc)	N	16	(ecc) ●
✓	Successful	PC0	Router0	ICMP	Light Green	16 (ecc)	N	16	(ecc) ●

Result:

The router configuration was successfully completed by connecting two switches to the router and attaching PCs to each switch. Appropriate IP addresses and subnet masks were configured for the router interfaces and the PCs. Connectivity was tested using the **ping** command from one PC to another PC in the same network and across different networks. The successful reply messages confirmed that the router was correctly configured and was able to route packets between the two networks, allowing proper communication between all connected devices.